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Upright Sitting Posture Detector based on Inertial Measurement Unit for Prolonged Desk Workers

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***Abstract*—Sitting for a long time in an improper posture may raise the risk of increasing spinal pain. Many factors are involved in creating lower back pain in people. One such main reason is the sedentary work style. Sitting upright is a good posture to reduce back-related pain. Several monitoring techniques are available to address the bad sitting posture problem. The proposed system is designed to monitor the posture of the person, and when it detects bad posture, it gives an alert message on the mobile. The system consists of an ESP 32 board, an Inertial Measurement Unit (IMU) sensor, an ultrasonic sensor, a T-shirt, and an Android mobile. The IMU was placed on the back side of the T-shirt based on the spinal cord's anatomical location. The bending angle will be assessed by the IMU sensor. The distance from the backrest of the chair is detected by an ultrasonic sensor. Based on the output from the IMU sensor and ultrasonic sensor, ESP 32 will process the data, send it over Bluetooth, and share it with a mobile app for continuous monitoring. The proposed method can help computer users and sedentary professionals actively achieve a straight sitting position for extended periods of desk work.**

Keywords—IMU Sensor, Ultrasonic Sensor, Posture Detection, Prolonged Sitting Posture.

I. Introduction

The most typical sign of spinal issues is low back pain. People at any age may experience back pain difficulties during their lives. A study [1] says 23% of people in the world might be affected by back pain issues. More than 80% of the population has experienced low back pain at least once [2]. In modern civilization, people spend a substantial amount of time sitting for work-related or relaxation activities [3],[4]. Long hours of sitting coupled with an abnormal stance could lead to an increased risk of spinal pain in sedentary employees [5]. Office employees typically maintain a seated position for the majority of their workday. According to studies, approximately 50 percent of office workers spend more than four hours per day seated.[6], [7], [8].

Sensor-based posture recognition may serve as a powerful tool for averting musculoskeletal disorders and enhancing the safety, health, and convenience of office workers [9]. In the last few years, some researchers and companies have begun to develop smart chair systems to track seated posture, while others have installed pressure sensors in chairs with ergonomic designs [10], [11]

This research is intended to provide a solution for back pain issues due to improper upright posture during desk jobs. The primary goal of the work is to design and develop a cost-effective sensory system to detect bad posture and give an alert when it detects it.

II. Materials and Methods

The proposed system has both a hardware section and a software design. The hardware parts used are the IMU sensor, ultrasonic sensor, ESP32, and battery. The software code is developed using the Arduino IDE.

A. Hardware Component Specifications

1) *Microcontroller:* The ESP 32 is a powerful micro-controller manufactured by Espressif Systems. It has so many attractive features, like wireless connectivity (built-in Wi-Fi and Bluetooth), a dual-core processor, low power consumption, a rich set of peripherals, integrated sensors, and various development Environment
2) *Inertial Measurement Unit:* The most widely used IMU sensor is the MPU-6050. It is a 6-axis IMU created by InvenSense. The IMU has a 3-axis gyroscope that calculates the speed at which each axis—yaw, pitch, and roll—rotates and a 3-axis accelerometer that monitors acceleration throughout every axis.
3) *Ultrasonic Sensor:* One popular ultrasonic distance sensor module is the HC-SR04. It is frequently utilized to measure distance. The ultrasonic echo-ranging mechanism underlies the operation of the HC-SR04. It generates ultrasonic pulses and timed the pulses' return to the sensor after bouncing off an object. The distance to the item is computed using this time measurement.
4) *Battery:* A 3.7-volt Li-ion rechargeable battery is used to power up the ESP-32 microcontroller board.
5) *Mobile:* An Android mobile device is used to connect with the sensing system to keep track of the sensing values and provide necessary alerts.

B. Software Component Specification

1) *Arduino IDE:* A software platform called the Arduino IDE (Integrated Development Environment) is used to program and upload code to ESP 32 microcontroller

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boards. This IDE is cross-platform. It includes many libraries (pre-written code) that can be used when interfacing with sensors and other peripheral devices. It also allows the installation of the required library packages using the library manager.

2) *MIT App Inventor:* MIT App Inventor: The Massachusetts Institute of Technology (MIT) created the web-based programming environment known as MIT App Inventor. Mobile applications can be created very easily by using a graphical block. MIT app uses Visual programming. No coding is required to build the app.

C. Hardware Connection setup

The two setups are designed to detect upright posture

1) *Ultrasonic sensor-integrated chair:* An ultrasonic sensor is connected to the ESP 32 microcontroller, which is powered by a 3.7-volt Li-ion rechargeable battery, as shown in Fig. 1. This unit is fixed to the top middle of the back side of the backrest in the chair, as portrayed in Fig.2

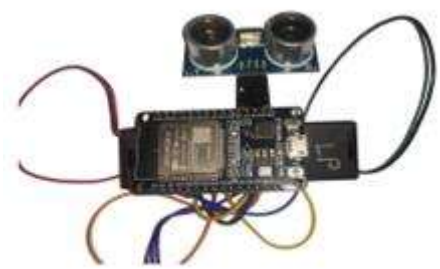

Fig. 1. ESP32 microcontroller with HC-SR04 sensor

Fig. 2. HC-SR04 Connected Chair

2) *IMU sensor-embedded T-shirt:* An IMU sensor is coupled to the ESP 32, which is power-driven by a rechargeable 3.7-volt Li-ion battery, as depicted in Fig.3. This part is attached to the upper back of the T-shirt, as shown in Fig.4.

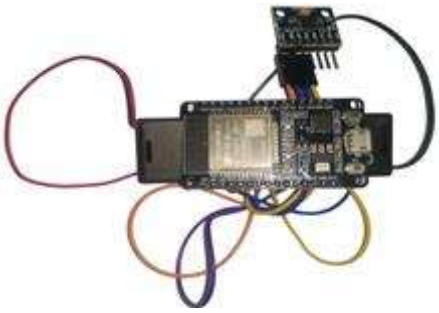

Fig. 3. ESP32 microcontroller with MPU 6050 sensor

Fig. 4. IMU sensor-embedded T-shirt

III. Results and Discussion

A Good healthy male (35 years) subject is used for the study. Consent is obtained from the participant. The purpose of the study, measures, and benefits are clearly explained. The selected subject has a work environment in which he has to spend an extended period in the chair. The distance between the edge of the ultrasonic sensor and the backrest is measured as shown in Fig.5 before the subject occupies the seat. The measured distance is 6 cm, and this value is set as the threshold. A good upright sitting posture can be maintained when the back of the subject is rested on the backrest of the chair. An Android application is designed using the MIT AppInventor to receive sensor data from the ESP 32 and display it. The application also shows an alert message when bad posture is detected. The workflow is denoted in the flow chart in Fig.6.

Fig. 5. Initial Distance measurement

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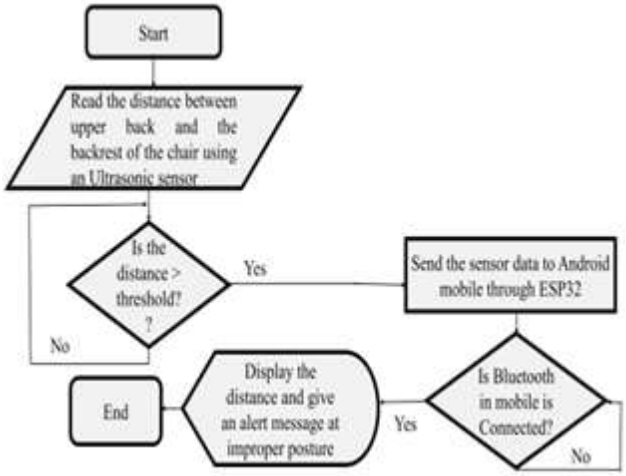

Fig. 6. Flow chart of Ultrasonic sensor distance measurement system

The subject is instructed to sit in the chair. The distance between the back of the subject and the backrest of the chair is measured by the sensory system. The sensor data is given to the ESP 32 controller. The program for reading sensor data and sending it to a mobile device is written and developed in the Arduino IDE and then loaded into the ESP32 controller. The microcontroller sends the distance value to the Android mobile device using Bluetooth wireless communication. Fig.7and Fig.8 shows the seating position of the subject during good and bad upright posture.

Fig. 7. Sitting at Normal Posture

Fig. 8. Improper posture detected using Ultrasonic sensor

An Android application is designed using the MIT AppInventor. Initially, after the application gets opened, the Bluetooth connection status is in disconnected mode. By pressing the connect Bluetooth button in the application, the available devices will be listed, as shown in Fig.9. From the list, the ESP32 control device has to be selected to connect that Bluetooth device to the mobile, and now the Bluetooth status is connected, as depicted in Fig.10.

Fig. 9. Bluetooth device selection

Fig. 10. Connection with ESP32 Bluetooth

When the received distance is equal to the threshold value, the posture is considered normal. The distance value is displayed in the application, as shown in Fig.11. Whenever the distance goes beyond the threshold value, it is categorized as improper posture, and an error message pops up on the mobile

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application, as shown in Fig.12. The output of the ultrasonic sensor-based approach is represented in Table.I. When the distance measured from the backrest is six cm,the posture is normal,so no alert message will be displayed, but when it is greater than six cm, the posture is bad, so the proper alert will be shown.

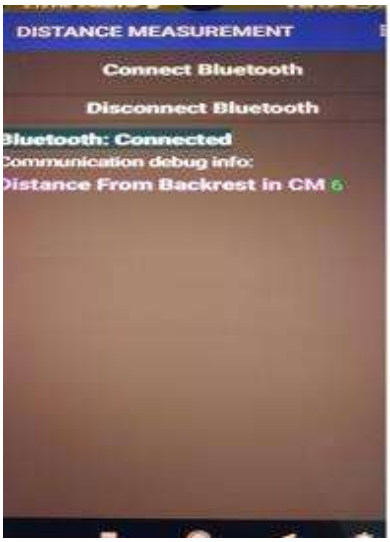

Fig. 11. Normal Posture

Fig. 12. Improper Posture alert message display

Hunching forward or slouching forward can cause the spine to curve and strain the lower back. In the next approach, the subject is requested to wear a T-shirt to which an IMU sensor unit is connected. The IMU sensor, along with the ESP32 microcontroller, is positioned around the thoracic spine area in the upper back. Fig.13 gives a brief overview of the workings of the IMU-integrated T-shirt. The subject is instructed to sit

TABLE I
ULTRASONIC SENSOR DISTANCE BASED POSTURE DETECTION

Test.No.	Distance from Backrest(cm)	Alert Message
1	6	-
2	7	Improper posture-Sit back
3	9	Improper posture-Sit back
4	18	Improper posture-Sit back

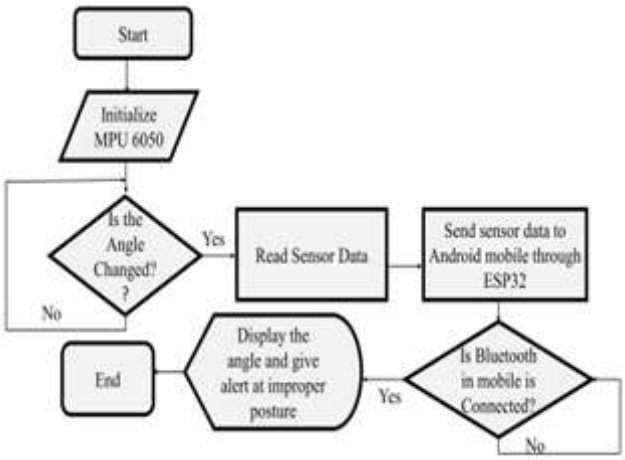

Fig. 13. Flow chart of IMU sensor angle measurement system

in the chair for a while. The change in the angle value of the upper back is monitored by the IMU sensor unit, and the corresponding data is given to the ESP32 microcontroller. Fig.14 and Fig.15 show the normal posture and improper upright posture of the subject, respectively.

Fig. 14. Sitting at a normal angle posture

The Arduino code is developed to read the IMU sensor value, and the angle is calculated using the sensor fusion algorithm. The angle value is sent to the Android mobile application. Fig.16 and Fig.17 show that the mobile application displays

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Fig. 15. Improper posture detected using IMU sensor

the error message for an angle value other than 90 degrees. The output of the IMU sensor-based method is shown in the Table.II. When the angle detected is 90 degrees, the posture is normal, so no alert message will be thrown. For other values of the sensor, an alert message will be given to the user.

TABLE II
IMU SENSOR ANGLE BASED POSTURE DETECTION

Test.No.	Angle (degrees)	Alert Message
1	75	Improper posture
2	90	-
3	97	Improper posture
4	128	Improper posture

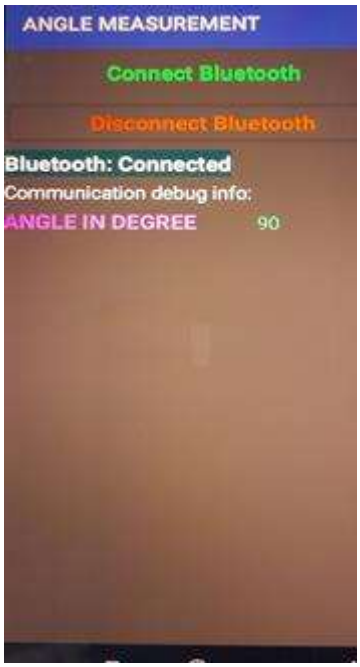

Fig. 16. Display angle value

Fig. 17. Improper posture alert

IV. CONCLUSION AND FUTURE WORK

The system is designed and developed using an IMU sensor, an ultrasonic sensor, and an ESP 32 as key hardware components. The system is capable of detecting improper upright posture using the ultrasonic sensor distance measurement method, and another approach is based on detecting variation in the angle of the upper back during slouching using an IMU sensor. An Android application is also being developed and interfaced with the system to keep track of the posture of the subject. The proposed system gives some breath to overcome the low back pain problems due to improper upright posture. Machine learning algorithms can be used in the future to find the different posture levels with more accuracy. This work is expected to be designed and implemented in the IoT-connected chair to provide more comfort.

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